$$B_2H_6 + 6CH_3OH \longrightarrow 2B(OCH_3)_3 + 6H_2$$

Hydroboration:

Diborane adds on to alkenes and alkynes in ether solvent at room temperature. This reaction is called hydroboration and is highly used in synthetic organic chemistry, especially for anti Markovnikov addition.

$$B_2H_6 + 6RCH=CHR \longrightarrow 2(RCH_2\text{-}CHR)_3B$$

Reaction with ionic hydrides

When treated with metal hydrides it forms metal borohydrides

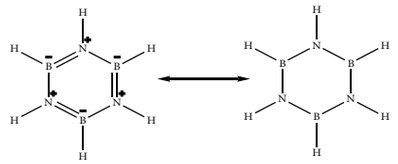
$$B_2H_6 + 2LiH \xrightarrow{\text{Ether}} 2LiBH_4$$
$$B_2H_6 + 2NaH \xrightarrow{\text{Diglyme}} 2NaBH_4$$

Reaction with ammonia:

When treated with excess ammonia at low temperatures diborane gives diboranediammonate. On heating at higher temperatures it gives borazole.

$$3B_2H_6 + 6NH_3 \xrightarrow{-153\ K} 3(B_2H_6.2NH_3) \text{ (or) } 3[BH_2(NH_3)_2]^+[BH_4]^-$$

$$3(B_2H_6.2NH_3) \xrightarrow[\text{Clossed vessel}]{\text{High temp}} 2B_3N_3H_6 + 12H_2$$



$2B_3N_3H_6$ (Borazole or Borazine - Inorganic benzene)

Structure of diborane:

In diborane two BH_2 units are linked by two bridged hydrogens. Therefore, it has eight B-H bonds. However, diborane has only 12 valence electrons and are not sufficient to form normal covalent bonds. The four terminal B-H bonds are normal covalent bonds (two centre - two electron bond or 2c-2e bond). The remaining four electrons have to be used for the bridged bonds. i.e. two three centred B-H-B bonds utilise two electrons each. Hence, these bonds are three centre- two electron bonds (3c-2e). The bridging hydrogen atoms are in a plane as shown in the figure 2.3. In diborane, the boron is sp^3 hybridised.

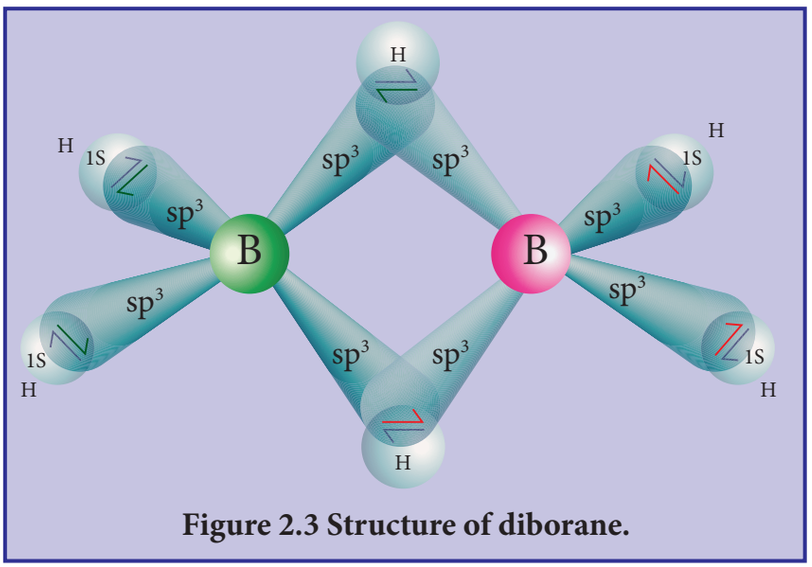


Figure 2.3 Structure of diborane.

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